Conveyor: Operation and Maintenance Manual

Applies to: Siemens SIMOTICS-C, Interroll DC-48, Dematic BeltMax. Document no. OM-CONVEYOR-2026.

1. Safety

Lock out and tag out all energy sources before any maintenance. Only trained personnel may work on this machine. Wear the PPE required by the plant safety rules.

2. Sensors and alarm thresholds

An alarm is raised when a value stays above the threshold for at least two consecutive samples. HIGH is a warning; HIGH_HIGH is critical.

Sensor	Location	Unit	Normal	HIGH (warning)	HIGH_HIGH (critical)
current	drive motor	A	approx. 9	> 14	> 18
temperature	gearbox	degC	approx. 48	> 70	> 85
vibration	drive end	mm/s	approx. 1.5	> 3.5	> 5.5

3. Alarm response

HIGH (warning): Acknowledge within 15 minutes. Review the trend of the last 2 hours and schedule an inspection within 24 hours. Follow the troubleshooting table for the sensor.

HIGH_HIGH (critical): Reduce load or stop the machine within 5 minutes and notify the shift lead. Open a corrective maintenance order. Do not restart until the root cause is found.

COMM_FAULT: Sensor readings are invalid (quality 'bad'). Check the sensor cable, connector and I/O module. If the fault lasts longer than 4 hours, take manual readings every hour.

4. Troubleshooting

4.1 High current (drive motor, A)

Probable causes:

- Material jam or overloaded belt
- Belt mistracking rubbing on the frame
- Seized carrying or return roller

Corrective actions:

1. Stop the conveyor and lock out before clearing any jam.
2. Check belt tracking and tension (deflection 15-20 mm per metre of span).
3. Locate seized rollers with a thermal camera and replace (SP-CNV-ROL-89).

4.2 High temperature (gearbox, degC)

Probable causes:

- Low gearbox oil level
- Wrong oil grade
- Continuous overload

Corrective actions:

1. Check gearbox oil level weekly.
2. Use only ISO VG 220 synthetic gear oil; change the oil every 2,500 operating hours.
3. At HIGH_HIGH stop the conveyor and check for overload before restart.

4.3 High vibration (drive end, mm/s)

Probable causes:

- Damaged belt splice
- Worn drive-end bearing
- Loose drive mounting bolts

Corrective actions:

1. Inspect the belt splice for lifting or cracks.
2. Replace the drive-end bearing unit SP-CNV-BRG-208.
3. Re-torque drive mounting bolts to 85 Nm.

5. Maintenance schedule

Interval	Task
Daily	Visual check of belt tracking
Every 30 days	Check belt tension and roller condition
Every 2,500 h	Change gearbox oil (ISO VG 220)
Yearly	Replace drive-end bearing unit SP-CNV-BRG-208